RSDS: A Specialized Loss Calculation Method for Dense Small Object Detection in Remote Sensing Images

Chengcheng Chen, *Graduate Student Member, IEEE*, Weiming Zeng, *Senior Member, IEEE*, Xiliang Zhang, *Graduate Student Member, IEEE*, Yuhao Zhou, Juan Yu, Yugang Chang, and Fei Wang

***Abstract*— Detecting dense small objects (DSOs) of varying scales still remains a challenging research problem in remote sensing imagery (RSI). Due to their weak feature extraction capabilities for small objects, most existing detection approaches struggle to handle the high proportion of DSO in RSI, thereby increasing the likelihood of missed detections. In addition, the close proximity and overlap of multiscale objects further complicate detection due to occlusion between bounding boxes. In this study, we systematically propose a novel loss function, remote sensing dense small target detection (RSDS), for detecting DSO in RSI, which contains three main components. The first is Gaussian reassignment loss (GRL), which adaptively redistributes sample weights to prevent any single sample (such as positive, negative, easy, and hard samples) from dominating the overall loss. To solve the zero-loss issue in traditional intersection over union (IoU) and intersection over ground truth (IoG) metrics when object boxes do not intersect, we design the Gaussian Wasserstein distance (WD) penalty loss, which models the eligible 2-D detection boxes as Gaussian distributions and calculates the similarity between them. The final one, which we called the occlusion box interaction loss, explains the attraction between DSO and the repulsion from their surroundings. Deploying RSDS not only significantly reduces the probability of missing DSO in RSI, but also enhances the detection accuracy in other similar computer vision tasks. Experiments on the HRSID, NWPU-10, and SSDD datasets show that some general models incorporating RSDS achieve precision improvements of 6.68%, 11.52%, and 5.26%, and 8.71%, 5.17%, and 9.34% in $mAP_{0.5}$ and $mAP_{0.5:0.95}$, respectively, compared with other baselines. The code will be found at https://github.com/CCC0090/RDSD-Loss.**

***Index Terms*— Convolutional neural networks, dense small objects (DSOs) detection, Gaussian Wasserstein distance (WD), loss function, occlusion and interaction, remote sensing imagery (RSI).**

Received 17 June 2024; revised 10 September 2024; accepted 7 October 2024. Date of publication 17 October 2024; date of current version 31 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 31870979 and in part by the 2023 Graduate Top Innovative Talents Training Program at Shanghai Maritime University under Grant 2023YBR013. *(Corresponding author: Weiming Zeng.)*

Chengcheng Chen, Weiming Zeng, Juan Yu, Yugang Chang, and Fei Wang are with the Digital Imaging and Intelligent Computing Laboratory, Shanghai Maritime University, Shanghai 201306, China (e-mail: shmtu_ccc@163.com; zengwm86@163.com; Yjuan007@163.com; 202340310001@shmtu.edu.cn; shine_wxf@163.com).

Xiliang Zhang is with the College of Information Engineering, Shanghai Maritime University, Shanghai 201306, China (e-mail: azxl1997@163.com).

Yuhao Zhou is with the College of Software Engineering, Tongji University, Shanghai 201804, China (e-mail: zhouyuhao@tongji.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3482358

I. Introduction

In RECENT years, remote sensing object detection has advanced rapidly, which has resulted in the development of many outstanding detection algorithms. However, one issue that is often overlooked in remote sensing imagery (RSI) containing multiscale objects is occlusion [1]. Occlusion can be categorized into two types: the first type is when an object is partially or completely obscured by other objects, and the second type is when detection boxes of dense small object (DSO) overlap each other. Due to the predominantly vertical shooting angle of RSI, the first type of occlusion is less common in optical remote sensing, particularly under favorable weather conditions. However, when adverse weather such as rain, snow, or haze occurs, the quality of optical RSI degrades significantly. This is especially true in spectral data [2], [3], where shorter wavelengths are more susceptible to interference from clouds and haze, which negatively impacts target recognition accuracy. In contrast, synthetic aperture radar (SAR) imagery offers greater resistance to interference, as it is unaffected by weather and lighting conditions, maintaining high-quality imaging in all-weather and complex environments. Nevertheless, SAR images face the second type of occlusion, particularly in multiscale target detection with densely packed small objects. When targets are closely spaced, occlusion between dense predicted bounding boxes reduces the alignment with ground truth (GT) boxes, increasing the risk of missed detections. This occlusion issue significantly impacts detection accuracy, particularly for small target detection, as illustrated in Fig. 1. Therefore, addressing dense target occlusion is crucial for improving the accuracy of remote sensing target detection.

In this study, we adopt the commonly used COCO definition [4] for object sizes in object detection, which defines sizes in terms of the number of pixels. According to this definition, small objects are those smaller than 32×32 pixels, medium-sized objects are those between 32×32 and 96×96 pixels, and large objects are those larger than 96×96 pixels. Dense objects are defined as those with overlapping GT boxes.

In machine learning and deep learning, loss functions [5], [6], [7], [8], [9], [10] are used to guide the training process of models. By minimizing the loss function, the predictions of the model become closer to the actual results, which improves the performance of the model. In multiscale remote

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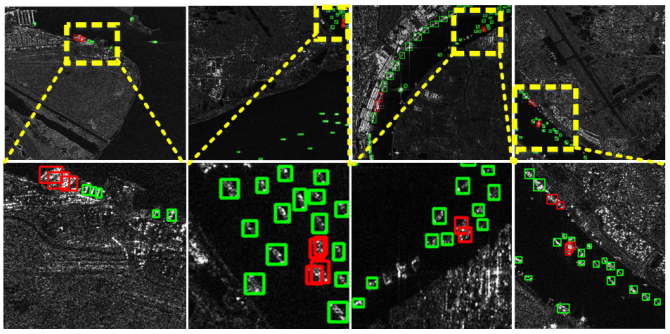


Fig. 1. Small objects and DSO diagram. Green boxes indicate small objects, while red boxes represent DSO.

sensing detection tasks, enhancing the accuracy of small object detection remains crucial. The two main factors influencing this critical technology are the model architecture [11], [12], [13], [14], [15], [16], [17] and the model loss [5], [6], [7], [10], [18], [19], [20]. As researchers continue to delve deeper into the field of object detection, numerous efficient detection algorithms based on statistical methods [21], [22], [23] and neural networks [15], [24], [25], [26], [27] have emerged. For object detection tasks involving known categories, our study focuses primarily on neural network-based approaches. These approaches exhibit strong feature extraction capabilities for medium and large-scale objects, which results in effective detection of these targets. However, as the targets become smaller and more densely packed, the performance of these detectors deteriorates. Therefore, in this study, we systematically construct a specialized loss function for DSO detection in multiscale remote sensing, particularly focusing on DSO detection tasks. This loss function is termed remote sensing dense small target detection (RSDS) loss.

The RSDS loss function first addresses the distribution of positive and negative samples and the allocation of hard and easy samples. An appropriate ratio of positive to negative samples can prevent the gradients [28] of the model from being dominated by negative samples during training, which improves the training effectiveness. The allocation strategy for hard and easy samples is crucial in object detection tasks given that it enhances the performance, generalization capability, and training efficiency of the model. This enhancement enables the model to better adapt to complex real-world scenarios. In this study, we design a sample-adaptive Gaussian assignment loss that uses Gaussian modeling to automatically partition positive and negative samples. More importantly, it resolves the issue of loss anomalies caused by the blurred boundaries between easy and hard samples. This resolution enables the model to learn from and optimize these samples during training, fully leveraging them to train the network.

Subsequently, for small object detection in remote sensing, traditional regression losses based on intersection over union (IoU) [29] and intersection over GT (IoG) [30] are sensitive to small objects, which leads to the discarding of many hard samples. This condition causes the network to be dominated by the loss from medium and large objects, which results in poor performance in small object detection tasks. We propose a Gaussian Wasserstein distance loss with regularization (RNWDloss) for small objects. This method employs a hybrid calculation approach, utilizing both WD and IoU metrics for small objects, while applying IoU-based loss computation for medium and large objects. In the calculation of the WD, detection boxes are initially modeled as 2-D Gaussian distributions. The similarity between these boxes is then measured using the WD. Finally, a penalty term is introduced to increase the regression loss weight for small objects, which accelerates their regression. On the one hand, this method overcomes the sensitivity of small objects to scale variations. On the other hand, it addresses the problem of nonoverlapping bounding boxes not contributing to the loss when measured by IoU by calculating the similarity between Gaussian distributions.

For DSO, we propose a bounding box interaction loss. This loss function can handle the complex scenario of occlusion among dense bounding boxes by adaptively calculating the interaction effects with surrounding anchors and GT boxes, centered on its anchor. Specifically, it increases the attraction between the predicted box and its corresponding GT box, while it enhances the repulsion between the predicted box and other predicted boxes and other GT boxes. This way ensures that the anchor more stably regresses to the correct position.

The main contributions of this study are as follows.

1) This study analyzes the proportion of large, medium, small, and DSO in the HRSID, SSDD, and NWPU-10 datasets, respectively. It quantitatively demonstrates that DSO is the primary affecting factor of detection accuracy.
2) To enhance the learning and classification capability of our model, we propose an adaptive Gaussian reassignment loss (GRL), which automatically calculates the threshold boundaries μ for positive and negative samples during training. The curve segment in Fig. 2 within $[\mu+\delta, \mu-\delta]$ is used as the weight mapping function for hard samples, which redistributes Gaussian weights to the samples.
3) We introduce a R_{NWD}Loss, a target size threshold filtering loss function that applies IoU and Gaussian Wasserstein metrics to targets of different scales. This approach not only achieves effective regression for medium and large targets but also addresses the sensitivity of small object anchors to positional changes (eliminating the zero-loss issue for nonoverlapping targets). In addition, the regularization design also reduces the risk of overfitting and improves the generalization capability of the model.
4) To ensure that predicted anchors of the DSO stabilize and accurately position themselves through effective interaction with the external environment, we propose a bounding box interaction loss ($\mathrm{Loss}_{\mathrm{inter}}$). This loss adaptively assigns weights between its own predicted and GT boxes and those of other targets during the training process.

II. Related Work

A. Object Detection and Label Assignment

Object detection, as a key task in computer vision, is typically categorized into two- [25], [31] and one-stage [11],

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[13], [15], [24], [27], [32], [33] detection algorithms based on the detection process. In both types of algorithms, the label assignment strategy is crucial. Its primary goal is to divide samples into positive samples (those that match the GT labels) and negative samples (those that do not match the GT labels). Then, it calculates the loss relative to the GT labels based on this division. This loss calculation guides the learning and optimization process of the model. This process directly impacts the performance and convergence speed of the model.

In object detection tasks, the reasonable allocation of positive and negative samples is crucial for maintaining training balance [16] and improving model performance. Typically, an algorithm predefines an IoU threshold: detection boxes with an IoU greater than this threshold with the GT labels are classified as positive samples, while those with an IoU less than the threshold are classified as negative samples. Commonly used loss functions, such as cross-entropy loss, focal loss [9], and smooth L1 loss [25], utilize these defined samples for training. However, the strategy for defining positive and negative samples directly affects model performance. Most methods use a clear-cut distinction between positive and negative samples, which can lead to an imbalance in sample distribution and ambiguity in distinguishing between hard and easy samples. Positive samples with high predicted probabilities and negative samples with low predicted probabilities are considered easy samples because these anchors are easily classified correctly. Conversely, positive samples with lower predicted probabilities and negative samples with higher predicted probabilities are hard samples, which are difficult to classify correctly. Traditional sample allocation strategies tend to result in an imbalanced ratio of positive to negative samples and blur the distinction between hard and easy samples. In such scenarios, the gradients of the model can be dominated by a large number of negative and easy samples, which hinders its ability to recognize hard samples. This limitation can lead to missed detections, which severely impacts the accuracy of the model.

In the field of object detection, strategies for defining positive and negative samples are diverse and can be broadly categorized into static and dynamic allocation strategies [34], [35], [36]. Static allocation strategies are among the most common methods for label assignment. For example, the online hard example mining (OHEM) [37] algorithm selects hard samples based on the loss of input samples. It automatically selects difficult samples for training via stochastic gradient descent. Although this method increases the weight of hard samples, it neglects the weight of easy samples. Focal loss [9] aims to further incorporate some beneficial easy examples into training without allowing the loss to be dominated by these simple samples. It introduces a weighting formula to down-weight easy examples, which ensures they contribute to the loss without dominating it. Gradient harmonizing mechanism (GHM) [38] also focuses on hard and easy samples similar to focal loss, but it attributes the imbalance between hard and easy samples to gradient imbalance. This approach posits that the gradients of certain particularly hard-to-classify samples (outliers) can be very high. If the model is forced to focus on these samples, then the accuracy of the model may be decreased.

However, static allocation strategies do not account for various sizes, shapes, or occlusion conditions of targets (GT). When a single anchor overlaps multiple GTs (e.g., in occluded frames), improper allocation can introduce harmful gradients to other GTs. OTA [39] views the label assignment problem as an optimal transport (OT) problem. It calculates the transport cost between GTs and all predicted boxes and utilizes global information to find the optimal one-to-many sample matching result. Slide loss [1] reduces hyperparameters by taking the average IoU value of all bounding boxes as the threshold μ; samples with IoU less than μ are considered negative, while those with IoU greater than μ are considered positive. Near the μ value, a proposed slide function weights the hard samples, which significantly reduces training time. The NWD-RKA strategy [40] significantly balances the optimization process across instances of varying scales by supplementing additional positive samples. Although these methods effectively address sample imbalance, they lack a clear definition for selecting hard samples and require manual setting of hard sample intervals. This limitation can lead to suboptimal performance in remote sensing small object detection tasks. This study proposes an adaptive GRL to address these issues. This loss calculates the Gaussian distribution of all samples adaptively based on different datasets. It then determines the range of hard samples according to the mean and variance of the Gaussian distribution, and it appropriately weights the losses of hard and easy samples.

B. Bounding Box Regression Loss

Regression loss is used to measure the discrepancy between the anchor box positions predicted by the object detection model and the GT. This loss not only helps the model more accurately locate the target objects but also guides the training process.

In bounding box regression loss, the issue with L1/L2 loss for coordinate regression lacking scale invariance was addressed by the introduction of IoU loss [29] in a study from ACM 2016. IoU loss treats the four coordinate points as a whole, which provides scale invariance. However, IoU loss fails to reflect the positional relationship between the anchor box and the GT when IoU is zero or a fixed value. To address this issue, GIoU [41] introduced the concept of the minimum enclosing rectangle (closure area) that covers the predicted and GT boxes. This approach not only focuses on the overlapping area but also considers other nonoverlapping regions. Thus, it better reflects the intersection situation of the two boxes within the closure area. However, when the predicted box is inside the GT box, GIoU degenerates to IoU, which results in failure to distinguish the relative positional relationship. Therefore, DIoU [5] introduces a penalty term based on the ratio of the center point distance to the diagonal distance. This insertion avoids optimization difficulties when the two boxes are far apart due to a large closure area. Nevertheless, if the center points of the two boxes coincide but the aspect ratios differ, then DIoU loss degenerates to

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IoU loss. To obtain more accurate predicted boxes, CIoU [5] adds an influence factor on top of DIoU, which considers the consistency of the aspect ratio between the predicted and GT boxes. Despite CIoU loss accounting for the overlap area, center point distance, and aspect ratio, the v term in its formula reflects the aspect ratio difference rather than the true difference between width and height and their confidence, which hinders effective model optimization in some cases. EIoU [6] builds upon CIoU by separating the aspect ratios to explicitly measure the differences in the three geometric factors: overlap area, center point, and side lengths. It also introduces focal loss to address the imbalance between easy and hard samples.

Traditional object detection loss functions do not initially consider the direction of mismatch between the predicted and GT boxes. This oversight leads to slower convergence and lower efficiency because the predicted box may "wander" during training, which ultimately results in a suboptimal model. The SIoU loss function [7] addresses this problem by considering the angle between the vectors of expected regression and redefines the penalty metric accordingly. In addition, traditional object detection loss functions primarily utilize the geometric relationship between boxes to enhance regression performance, which often neglects the impact of sample difficulty distribution on bounding box regression. Focaler-IoU [10] improves detector performance across various detection tasks by focusing on regression samples of varying difficulty.

In small object detection in remote sensing, using IoU as a metric for regression loss can lead to varying sensitivities of anchor boxes at different scales. As a result, networks are more favorable to medium to large objects and potentially miss small or tiny targets. To address this issue, Wang et al. [8] proposed a new metric that measures the similarity of bounding boxes using WD instead of the standard IoU. The proposed normalized Wasserstein distance (NWD) is utilized to measure the similarity of derived Gaussian distributions. The primary advantage of WD is its ability to measure distribution similarity in a probabilistic manner [42], which ensures that nonoverlapping targets also incur a loss. Therefore, NWD is insensitive to objects of different scales, which makes it more suitable for measuring similarity between tiny objects. This loss achieves optimal performance in small object detection. SIWD [43] introduces the concept of scale invariance, making the network loss function more accommodating to multiscale targets. However, in complex environments with multiscale target aggregation, bounding boxes are susceptible to the influence of surrounding overlapping boxes during regression, which leads to the "wandering" issue of predicted boxes. To address this concern, regularization of the loss on probability distributions is performed, and a penalty term is added to enforce box regression.

C. Dense Remote Sensing Small Target Occlusion

RSI, captured by distant sensors on satellites, airplanes, and other platforms, provides comprehensive global perspectives and covers extensive areas. Due to the resolution limitations of these sensors, the targets in RSI often exhibit multiscale features, and their bounding boxes frequently overlap. This presents challenges for small object detection, as small objects occupy minimal space and the lack of resolution makes it difficult to capture details. Moreover, in cases of target clustering, the regression process of anchor boxes can be significantly influenced by other neighboring anchor boxes and bounding boxes.

This phenomenon has two main impacts on model accuracy. Initially, when DSO are present, the predicted bounding boxes may be influenced by postprocessing nonmaximum suppression (NMS) operations, which leads to the filtering out of a large number of correctly predicted bounding boxes and results in missed detections. Subsequently, the proximity of bounding boxes between small targets in clustered regions may cause interference from surrounding occluding boxes during anchor regression. In such cases, the matching capability between anchors and GT bounding boxes may weaken. In addition, occluding boxes and GT bounding boxes of surrounding objects can also negatively impact the regression between the anchors and the GT bounding boxes themselves. These factors result in the phenomenon of "wandering" movement of anchors near the correct GT bounding boxes, or even deviation from the correct position. Ultimately, this condition causes difficulty for anchors to regress accurately, which leads to challenges in model convergence.

The repulsion loss [30] employs a weight allocation mechanism to enhance the attraction between the predicted bounding box and its corresponding GT box while simultaneously reducing the attraction between the predicted bounding box and the bounding boxes of other objects. Given that this scenario is based on IoU regression, even when using methods like IOG to enhance the attraction between the predicted bounding box and the GT box, in the case of clustered small targets, if the predicted bounding box and the GT box become separated, then it still results in an IoU of 0. Consequently, the probability of missing small targets increases.

Therefore, in the current stage of DSO detection, the key issue is enabling anchors to robustly regress to the correct positions in complex environments with clustered small objects.

III. Method

The RSDS loss proposed in this study, represented by L in (1), primarily consists of four components: object box classification loss, confidence loss, regression loss, and interaction loss. On the one hand, the RSDS loss addresses the issue of imbalanced weighting in assigning positive and negative samples, especially simple and difficult samples, in DSO detection. On the other hand, it resolves the sensitivity of IoU to the shape and position of small or extremely small objects. Moreover, it addresses the regression problem of anchors under DSO occlusion scenarios

$$L = \text{Loss}_{\text{cls}}(\text{GR}) + \text{Loss}_{\text{obj}}(\text{GR}) + R_{\text{NWD}} + \text{Loss}_{\text{inter}}. \quad (1)$$

In the following sections, Section III-A provides a detailed description of the Gaussian sample reallocation method applied to classification loss and confidence loss. Section III-B introduces the R_{NWD}Loss. Section III-C elaborates the interaction loss for DSO scenarios.

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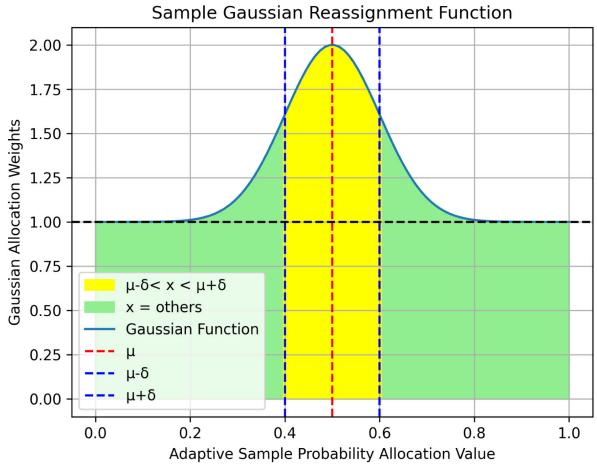


Fig. 2. Sample Gaussian reassignment function. μ represents the mean of the Gaussian distribution, and δ is a hyperparameter.

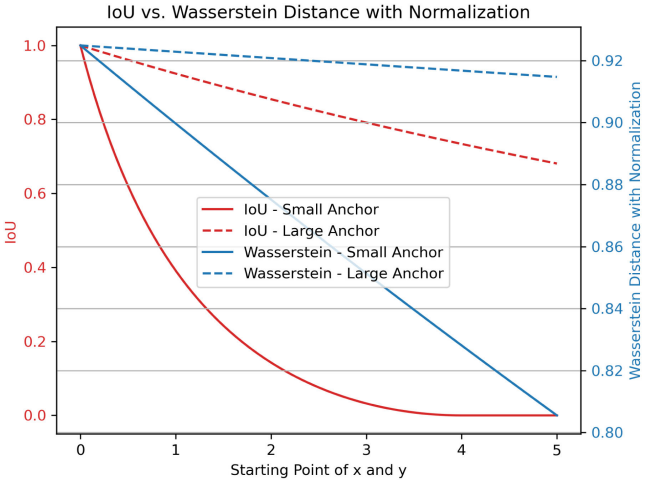


Fig. 3. Comparison of IoU and WD with normalized calculation results under normal conditions.

A. Method for Addressing Sample Imbalance

The issue of sample imbalance in object detection is a significant affecting factor of model performance. Typically, no clear standard exists for the division of positive and negative samples, and the thresholds for this division are often preset based on empirical knowledge. This problem affects the ability of the model to balance positive and negative samples and its generalization performance. In this study, we propose the GRL from the perspectives of IoU and probability, as illustrated in Fig. 2.

The core idea of GRL is to adaptively adjust the sample weights in the loss function using a Gaussian distribution, thereby more effectively addressing the issue of positive and negative sample classification. The method involves the following steps: first, the IoU values between the GT boxes and the predicted boxes are calculated. Next, these values are used to compute the mean and variance of a Gaussian distribution, and an adaptive Gaussian probability density function is constructed, as illustrated by the blue curve in Fig. 2. The computation process is detailed in (2). Then, the IoU values of each sample are input into the Gaussian function to obtain their corresponding values within the distribution. In the next step, the mean (μ) of the Gaussian distribution serves as the threshold for distinguishing between positive and negative samples, with the interval [0, μ] representing negative samples and [μ, 1] representing positive samples. We apply weighting to the hard samples to further prevent the loss gradient from being dominated by easy samples. By introducing $\mu + \delta$ and $\mu - \delta$, the Gaussian distribution is divided into three segments, with the interval [$\mu - \delta$, $\mu + \delta$] corresponding to hard samples. Finally, we employ (3) as the weighting function for the GRL, which assigns appropriate weights to hard samples based on the Gaussian function values. This approach ensures greater focus on hard-to-classify and misclassified examples.

The GRL addresses the issue of ambiguous positive and negative sample classification by adaptively setting the threshold and assigning higher weights to hard samples. Through these strategies, the GRL enables a more balanced handling of positive and negative sample distributions, while improving the robustness and generalization ability of the model across various complex scenarios

$$p\left(x \mid \mu_k, \sigma_k^2\right) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right) \tag{2}$$

$$\omega(x) = \frac{p\left(x \mid \mu_k, \sigma_k^2\right)}{\sum_{k=1}^{K} p\left(x \mid \mu_k, \sigma_k^2\right)} + 1 \tag{3}$$

where $p(x \mid \mu_k, \sigma_k^2)$ denotes the probability of sample x being assigned to the kth Gaussian distribution. This represents K Gaussian distributions, each following $N(\mu_k, \sigma_k^2)$.

B. Multiscale R_{NWD}Loss Calculation Method

This study introduces the multiscale object R_{NWD}Loss, which combines the advantages of IoU and WD loss calculation methods. This loss function is applied to the calculation of losses for targets of different scales in remote sensing detection, achieving optimal results.

In traditional IoU thresholds, the performance for small and minuscule targets has been subpar when calculating the loss between predicted and GT bounding boxes. When the overlap area between the GT and predicted boxes is extremely small, IoU values fluctuate significantly and even reach zero. In such scenarios, IoU regression losses may impede model convergence. In addition, the IoU function exhibits discontinuities and nondifferentiability at the boundaries. This condition not only leads to model entrapment in local optima but also hinders gradient updates, which affects training efficacy.

Currently, a method has emerged that evaluates the loss generated between anchors by computing the similarity of probability distributions. This approach addresses the issue of discontinuity and nondifferentiability at the boundaries encountered during anchor regression. It also handles regression for small targets more effectively. Fig. 3 compares the computation results of IoU and WD. The figure demonstrates how WD overcomes the drawback of IoU yielding 0 in nonintersecting scenarios, but it exhibits variations in effectiveness for targets of different sizes. As shown in the figure, WD is

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more accommodating to small-sized targets. However, Fig. 2 also highlights the drawbacks of the WD. Compared to IoU, it tends to produce larger losses, making the model more difficult to converge.

To address the aforementioned issues, this study combines the WD [8], [44], [45] and IoU [5], [6], [7], [10] methods to calculate more accurate loss values. The former models the anchors of small objects as 2-D Gaussian distributions and uses the WD to compute the loss. This approach mitigates the sensitivity of traditional IoU to position changes of small objects. Specifically, modeling the anchors of small objects as 2-D Gaussian distributions involves the following three steps.

1) *Anchor Representation Selection and Computation of Anchor Statistical Features:* Here, we represent anchors as (X, Y, W, H), which denote the coordinates of the top-left corner of the box along with its width and height. We also compute statistical features for each anchor, such as its centroid coordinates, width, and height.
2) *Modeling the Inscribed Ellipse of Rectangles:* Based on the statistical features of anchors, we model the density contours of 2-D Gaussian distributions, as depicted in Fig. 4. This modeling aims to better capture variations in the position and shape of the target box.
3) *Modeling a 2-D Gaussian Distribution:* Define the probability density function of the 2-D Gaussian distribution based on the statistical characteristics and covariance of the anchor.

Based on the representation of anchors, we first transform the coordinates of the top-left corner (X, Y) to the center coordinates (C_x, C_y) of the rectangle using the following equation:

$$\begin{aligned} C_x &= X + W/2 \\ C_y &= Y + H/2. \end{aligned} \tag{4}$$

Subsequently, each anchor is modeled with an inscribed ellipse based on (5), where (x, y) represents points on the ellipse, and $W/2$ and $H/2$ denote half the width and height of the ellipse, respectively,

$$\frac{(x - C_x)^2}{\left(\frac{W}{2}\right)^2} + \frac{\left(y - C_y\right)^2}{\left(\frac{H}{2}\right)^2} = 1. \tag{5}$$

Then, according to (6), we compute the covariance matrix Σ. In this scenario, the covariance matrix Σ is a diagonal matrix with elements on the diagonal representing the variances of the width and height of the rectangle. We assume independence between the width and height of the rectangle; thus, the off-diagonal elements of the covariance matrix are zero ($\rho = 0$)

$$\Sigma = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix} \tag{6}$$

where σ_x^2 and σ_y^2 represent the variances of the width (W) and height (H) of the anchor, respectively, while ρ denotes the correlation coefficient between them.

Finally, the probability density function $p(x, y)$ in the process of modeling the anchor as a 2-D Gaussian distribution can be expressed as follows:

$$p(x, y) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2}(X - \mu)^{\mathrm{T}}\Sigma^{-1}(X - \mu)\right) \tag{7}$$

where $\mathbf{X} = \binom{x}{y}$, $\mu = \binom{C_x}{C_y}$, $\Sigma = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}$, and $\det(\Sigma)$ is the determinant of the covariance matrix Σ.

This method transforms the differences between bounding boxes into differences between two Gaussian distributions, utilizing a 2-D Gaussian distribution to calculate the regression loss for small objects. Fig. 4(c) illustrates the 3-D Gaussian distribution of small objects, demonstrating that the distribution range of small objects extends to resemble that of medium and large objects, thereby reducing the differences between them.

C. Method for Calculating the WD Between Two Gaussian Distributions

For two multivariate Gaussian distributions, where X follows $\mathcal{N}(\mu_1, \Sigma_1)$ and Y follows $\mathcal{N}(\mu_2, \Sigma_2)$, their second-order WD in this study can be defined as the equation: $W_2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) \triangleq \inf \mathbb{E}\left(\|X - Y\|_2^2\right)^{1/2}$. This equation specifies that the definition is the minimum expectation among all possible couplings of the two distributions (coupling refers to all possible ways of matching the random variables of the two distributions). X and Y represent samples drawn from the two distributions, respectively, and $\|X - Y\|_2^2$ denotes the squared Euclidean distance (ED) between these samples. Therefore, the WD between the two normal distributions is further calculated using the following equation:

$$\begin{aligned} &W_2^2(\mathrm{N}(\mu_1, \Sigma_1), N(\mu_2, \Sigma_2)) \\ &\quad = \|\mu_1 - \mu_2\|^2 + \operatorname{trace}\left(\Sigma_1 + \Sigma_2 - 2\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right). \end{aligned} \tag{8}$$

In this context, μ_1 and μ_2 are the mean vectors of two normal distributions, while Σ_1 and Σ_2 represent their respective covariance matrices. For Gaussian distributions, the covariance matrices Σ_1 and Σ_2 must be symmetric. Therefore, (8) can be further derived into (9), the derivation process of which is detailed in Appendix A

$$W_2^2 = \|\mu_1 - \mu_2\|^2 + \left\|\Sigma_1^{1/2} - \Sigma_2^{1/2}\right\|_{\text{Frobenius}}^2. \tag{9}$$

To reduce computational complexity, this study proposes a simplified version of similarity calculation based on (9), as shown in (10): the square of the WD between two distributions can be represented by the square of the ED between their means, along with the difference between their covariance matrices. This is consistent with the elements involved in the square of the ED between two anchor vectors in object detection, which encompass differences between their center points and differences between their widths and heights

$$W_2^2(N_a, N_b) = \|V_a - V_b\|_2^2 \tag{10}$$

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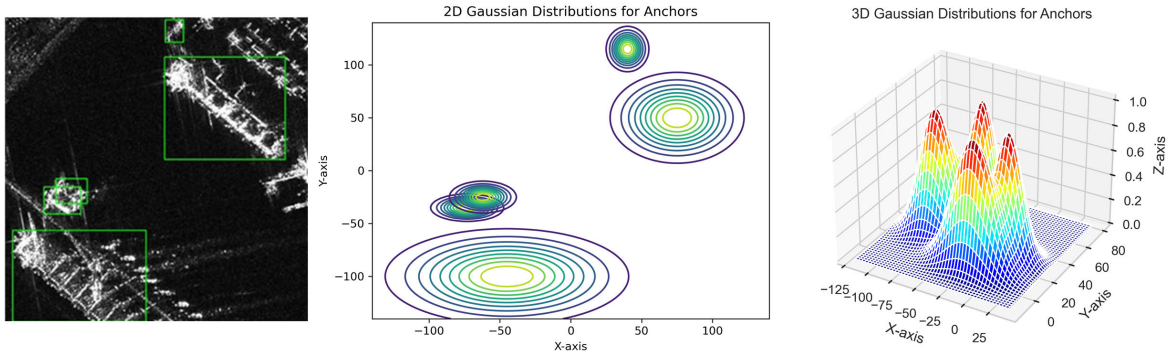


Fig. 4. Visualization of anchor modeling. (a) GT image. (b) Two-dimensional Gaussian density contour map of the anchor. (c) Three-dimensional Gaussian distribution map of the anchor.

where $V_a = [C_{x_a}, C_{y_a}, (W_a/2), (H_a/2)]$ and $V_b = [C_{x_b}, C_{y_b}, (W_b/2), (H_b/2)]$ represent vectors containing information on the center coordinates, width, and height. The derivation of the ED equation between vectors and the simplified algorithm flow are detailed in Appendixes B and C, respectively.

D. Robustness and Nonsmoothness Issues in Multiscale Object Distributions

Although we have modeled the anchor boxes of small objects as Gaussian distributions and employed a hybrid loss to calculate the loss values, the loss for medium and large-scale objects can be accurately computed using IoU. However, excessive use of the WD for small objects can reduce model accuracy and increase training time. To address this, we designed an anchor filter that routes small objects below a certain threshold to the hybrid loss calculation path. Meanwhile, medium and large objects are directed to the IoU-based calculation path. Thus, the R_{NWD}Loss can calculate optimal loss values for multiscale targets. However, this multiloss overlay often introduces noise and outliers, which results in a nonsmooth loss function. To address this issue, we initially devised an adaptive regularization term, as shown in (11), and incorporated it into (12), followed by application to WD computation. This adaptation effectively controls the smoothness of distributions and the magnitude of model parameters. As a result, the stability and generalization capability of the model is enhanced

$$\text{reg}_{\text{term}} = \lambda\left(\sum_{i=2}^{3}\left(\sum_{j=1}^{N}|\text{pred}[j,i]| + \sum_{j=1}^{N}|\text{gt}[j,i]|\right)\right) \tag{11}$$

$$\text{reg}_{\text{value}} = W_2^2(N(\mu_1, \Sigma_1), N(\mu_2, \Sigma_2)) + \text{reg}_{\text{term}} \tag{12}$$

where $W_2^2(\text{N}(\mu_1, \Sigma_1), N(\mu_2, \Sigma_2))$ is derived from (9), and λ is a regularization coefficient used to control the degree of regularization. N denotes the number of samples. The terms $\sum_{i=2}^{3}\sum_{j=1}^{N}|\text{pred}[j,i]|$ and $\sum_{i=2}^{3}\sum_{j=1}^{N}|\text{gt}[j,i]|$ represent the width and height values in the predictions and GT, respectively.

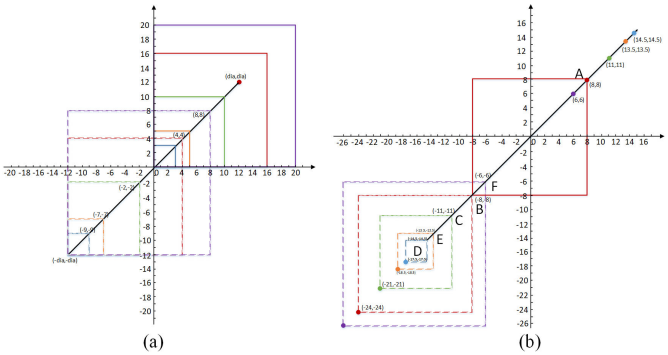


Fig. 5. Comparison of sensitivity between the R_{NWD} metric and the IoU metric regarding object scale and position. (a) Process of five anchors of different scales moving from the bottom-left corner (-dia, -dia) to (dia, dia). (b) Process of five anchors at different positions moving from the bottom-left corner to their respective colored points, with each movement representing a one-pixel shift.

Next, we process the $\text{reg}_{\text{value}}$ in exponential form to produce a normalized similarity metric, as follows:

$$R_{\text{NWD}} = \exp\left(-\frac{\sqrt{\text{reg}_{\text{value}}}}{C}\right) \tag{13}$$

where C is a constant related to the dataset model, set to 12.8 in this study.

Finally, R_{NWD}Loss generates the loss value according to the configuration in (14), where R represents $R_{\text{NWD}}\text{Loss}(N_a, \text{N}_b)$

$$R = \begin{cases} 1-\varphi, & \text{size} \geq \theta \\ \eta(1-R_{\text{NWD}}) + \gamma(1-\varphi), & \text{size} < \theta \end{cases} \tag{14}$$

where $\eta \in [0, 1]$ and $\gamma \in [0, 1]$ are hyperparameters such that $\eta + \gamma = 1$; φ denotes the IoU-based loss value; size refers to the area of the object; and θ is the area threshold.

To further demonstrate the robustness of the proposed R_{NWD} metric compared to IoU-based measures regarding small target scale and positional variations, further analysis was conducted. In accordance with the requirements of Fig. 5, we present deviation curves of IoU and R_{NWD} regarding scale and positional variations, as shown in Fig. 6.

Based on the above experimental analysis, it has been demonstrated that the proposed R_{NWD} exhibits greater

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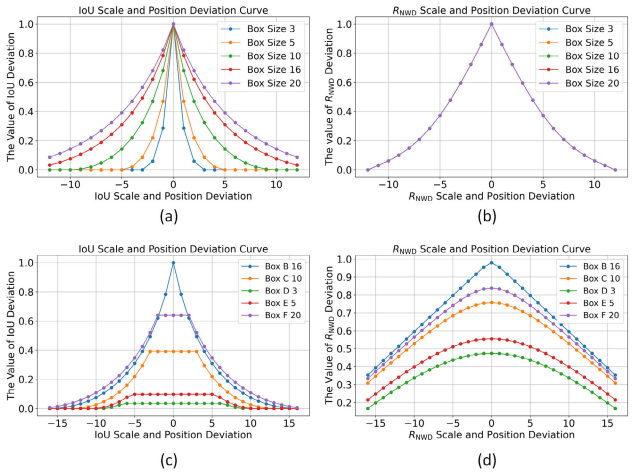


Fig. 6. Comparison of deviation curves for IoU and R_{NWD} regarding scale and position. According to the setup in Fig. 5(a), the deviation curves for IoU and R_{NWD} are shown in (a) and (b), respectively. Based on the setup in Fig. 5(b), the deviation curves for IoU and R_{NWD} are presented in (c) and (d), respectively.

robustness to scale and position variations in small targets. Finally, to more clearly illustrate the logic of R_{NWD}Loss, Algorithm 1 outlines its calculation process.

E. Method for Addressing Precise Localization of Densely Occluded Targets

In this study, the NWD regression loss is used to optimize the overlap between predicted boxes and GT boxes. However, as shown in Fig. 7(a), dense occlusion problems may arise. The mutual occlusion among densely packed targets can compromise the accuracy of the regression, causing predicted boxes to be affected by neighboring targets, leading to suboptimal performance, as illustrated in Fig. 7(b). During the regression process, the green bounding box is easily influenced by the predicted box and GT box of adjacent target B, causing it to deviate from the correct GT box A. If the bounding box lies between A and B, and the distance from the correct GT box exceeds the NMS threshold, it can result in inaccurate detections. Conversely, if the distance is below the NMS threshold, the risk of missed detections increases.

To address these issues, this study proposes an interaction loss for densely packed small targets [as defined in (15)]. The interactive loss operates independently and is computed in parallel with confidence and classification losses. After applying the R_{NWD}Loss, interactive loss further optimizes the positions of predicted and GT boxes, effectively mitigating interference between different target boxes and improving the detection accuracy for densely packed small targets. A comparison of the heatmaps in Fig. 7(c) and (d) demonstrates that, with the inclusion of interactive loss, densely packed targets are no longer represented as a single region. Instead, the targets are more finely segmented, with clear improvements in the internal details of the yellow bounding box region

$$\mathrm{Loss}_{\mathrm{inter}} = L_{\mathrm{At}} + \alpha L_{\mathrm{OpGT}} + \beta L_{\mathrm{OpBox}}. \tag{15}$$

The interactive loss consists of three components: L_{At}, which is the attraction loss between the anchor and its own

Algorithm 1 R_{NWD}Loss

Require: pred, target, $\epsilon = 1e{-}7$, constant $= 12.8$, $\mathrm{reg}_{\mathrm{weight}} = 0.1$

Ensure: R_{NWD}Loss

Compute anchor areas: $\mathrm{anchor}_{\mathrm{area}} = \mathrm{pred}[:, 2] \times \mathrm{pred}[:, 3]$

Set filter aperture size: $\mathrm{valid}_{\mathrm{indices}} = \mathrm{anchor}_{\mathrm{area}} < 32 \times 32$

if any $\mathrm{valid}_{\mathrm{indices}}$ **then**

  Extract mean matrices from predictions and targets: $\mathrm{pred}_{\mathrm{mean}} = \mathrm{pred}[:, :2]$, $\mathrm{target}_{\mathrm{mean}} = \mathrm{target}[:, :2]$

  Extract covariance matrices from predictions and targets:

  $\mathrm{pred}_{\mathrm{cov}} = \mathrm{torch.diag}_{\mathrm{embed}}(\mathrm{pred}[:, 2:4].\mathrm{float}(), \mathrm{dim1} = -2, \mathrm{dim2} = -1)$

  $\mathrm{target}_{\mathrm{cov}} = \mathrm{torch.diag}_{\mathrm{embed}}(\mathrm{target}[:, 2:4].\mathrm{float}(), \mathrm{dim1} = -2, \mathrm{dim2} = -1)$

  Construct distributions: $\mathrm{pred}_{\mathrm{dist}}$, $\mathrm{target}_{\mathrm{dist}}$:

  $\mathrm{pred}_{\mathrm{dist}} = \mathrm{MultivariateNormal}(\mathrm{pred}_{\mathrm{mean}}, \mathrm{pred}_{\mathrm{cov}})$

  $\mathrm{target}_{\mathrm{dist}} = \mathrm{MultivariateNormal}(\mathrm{target}_{\mathrm{mean}}, \mathrm{target}_{\mathrm{cov}})$

  Compute Wasserstein distance: WD

  $\mathrm{WD} = \sqrt{\sum(\mathrm{pred}_{\mathrm{dist}}.\mathrm{mean} - \mathrm{target}_{\mathrm{dist}}.\mathrm{mean})^2}$

  $+ \sum \sqrt{\mathrm{torch.linalg.det}(\mathrm{pred}_{\mathrm{dist}}.\mathrm{cov}_{\mathrm{mat}} + \epsilon)}$

  $+ \sum \sqrt{\mathrm{torch.linalg.det}(\mathrm{target}_{\mathrm{dist}}.\mathrm{cov}_{\mathrm{mat}} + \epsilon)}$

  Compute regularization term: $\mathrm{reg}_{\mathrm{term}}$

  $\mathrm{reg}_{\mathrm{term}} = \mathrm{reg}_{\mathrm{weight}} \times (\sum |\mathrm{pred}[:, 2:4]| + \sum |\mathrm{target}[:, 2:4]|)$

  Compute R_{NWD}:

  $R_{\mathrm{NWD}} = \exp\left(-\frac{\mathrm{WD} + \mathrm{reg}_{\mathrm{term}}}{\mathrm{constant}}\right)$

else

  return IoU

end if

Update lbox:

$\mathrm{lbox} += \gamma \times (1.0 - \mathrm{IoU.mean}()) + \eta \times (1.0 - R_{\mathrm{NWD}}.\mathrm{mean}())$

GT; αL_{OpGT}, which is the opposition loss between the anchor and surrounding targets; and βL_{OpBox}, which is the opposition loss between the anchor and surrounding target anchors. α and β are hyperparameters used to balance the weights of different loss terms.

The L_{At} loss component ensures stable regression of the anchor toward its own GT in specific scenarios. We compute the loss value between the anchor and its own GT using the exponential normalization value in (13). Compared with IoU, this approach better considers the characteristics of small targets. Therefore, the resulting attraction value is calculated using the following equation:

$$L_{\mathrm{At}} = \frac{\Sigma_{P \in \mathcal{P}^+} R_{\mathrm{NWD}}\left(B^P, G^P\right)}{|\mathcal{P}^+|} \tag{16}$$

where $|\mathcal{P}^+|$ represents the cardinality of the set of all positive samples, i.e., the total number of samples. B^P and G^P, respectively, denote the regression box and GT box. The role of the L_{OpGT} loss is to minimize the influence of surrounding GT on its own regression anchor, as detailed in the following

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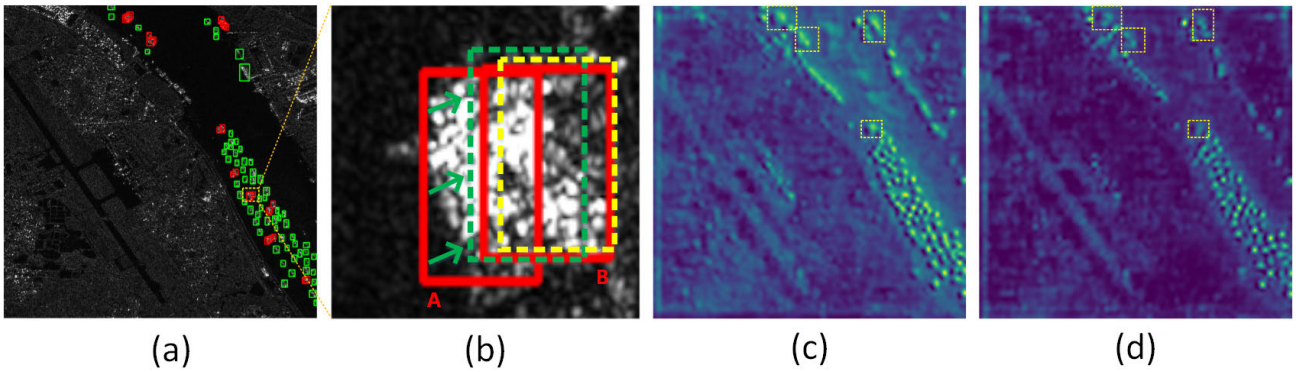


Fig. 7. Anomaly chart for bounding box localization in DSOs. (a) Original image, where the red boxes represent densely occluded target areas and the green boxes represent normal bounding boxes. In (b), the red bounding boxes A and B are GT boxes of the same class, while the green and yellow anchors represent the predicted boxes for A and B, respectively (using two densely occluded targets as examples, assuming each target has only one predicted box). (c) and (d) Heatmaps for cases without and with the addition of $\text{Loss}_{\text{inter}}$, respectively.

equation:

$$L_{\text{OpGT}} = \frac{\Sigma_{P\in\mathcal{P}^+}\text{Smooth}^*\left(R_{\text{NWD}}\left(B^P, G^P_{\text{Sec}}\right)\right)}{|\mathcal{P}^+|}. \tag{17}$$

In this equation, the Smooth* function designed in this study serves as a segmented function that reflects the importance of opposition loss, as demonstrated in (18). It exhibits superior repulsion effects to $\text{Smooth}_{\ln}$, given that it exerts stronger constraints particularly effective for dense small target regression. The comparison of the two constraint functions is visually depicted in Fig. 8. In (17), G^P_{Sec} refers to the GT box with the maximum R_{NWD} in the remaining GT set, aside from the matching GT box G^P for predicted box P, as specified in (19). Visualizing (18) in Fig. 8 reveals that, as the R_{NWD} between predicted box P and surrounding target boxes G increases, the resulting loss also amplifies. Thus, this mechanism effectively prevents predicted boxes from shifting toward GT boxes of surrounding targets

$$\text{Smooth}^* = \begin{cases} \ln\left(\dfrac{1+x}{1-x}\right), & x \le t \\ \dfrac{2(x-t)}{1-t^2} + \ln\left(\dfrac{1+t}{1-t}\right), & x > t \end{cases} \tag{18}$$

$$G^P_{\text{Sec}} = \arg\max\ R_{\text{NWD}}(G, P), \quad G \in \mathcal{G}\backslash\left\{G^P\right\}. \tag{19}$$

LOpBox induces opposing forces between neighboring predicted boxes p_i and predicted boxes p_j of other surrounding targets. This way ensures that p_i and p_j are not influenced by other predicted boxes during regression, which allows them to approach their respective correct GT boxes independently. Its definition is outlined in (20). Based on (20), we observe that, as the R_{NWD} between predicted box p_i and surrounding predicted boxes p_j increases, the resulting loss also escalates. Fig. 8 vividly illustrates the effectiveness of Smooth*, which effectively prevents two predicted boxes from being filtered out by NMS due to their proximity and consequently reduces the misdetection rate of the model

$$L_{\text{OpBox}} = \frac{\sum_{i\neq j}\text{smooth}^*\left(R_{\text{NWD}}\left(B^{P_i, B^{P_j}}\right)\right)}{\sum_{i\neq j}1\left[R_{\text{NWD}}\left(B^{P_i, B^{P_j}}\right) > \theta\right] + \varepsilon}. \tag{20}$$

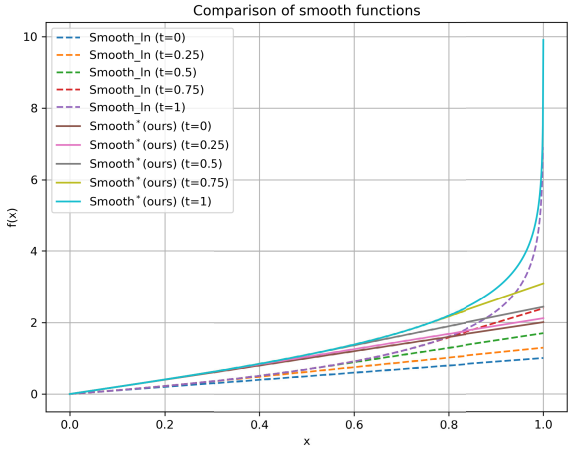


Fig. 8. Comparison chart of constraint function effects. Here, $t \in [0, 1]$ is the smoothing parameter. As t decreases, the function becomes less sensitive to outliers.

IV. Experiment

The experimental section is organized as follows. In Section IV-A, we conduct an analysis of target size and density on three publicly available datasets, namely, SSDD [46], HRSID [47], and NWPU-10 [48], to validate the prerequisites for the experiments. Section IV-B introduces the experimental setup and evaluation metrics required for the experiments. In Section IV-C, we perform ablation and comparative experiments of detection models on the three datasets, followed by comprehensive analyses from various perspectives. In Section IV-D, we compare several state-of-the-art object detection models with the model incorporating the RSDS loss, which validates the effectiveness of the RSDS loss from multiple angles.

A. Dataset and Data Analysis

This study employs three widely used public datasets, namely, HRSID [47], SSDD [46], and NWPU-10 [48]. The specific parameters of the datasets are outlined in Table I.

To validate the effectiveness of the RSDS loss in the context of remote sensing dense small targets in the subsequent

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TABLE I
DATASET PROPERTIES TABLE

Dataset	Data Type	Source	Amount	Resolution	Polarization mode
SSDD [46]	SAR	RadarSat2 TerraSARX Sentinel1	1160	1m-15m	HH,HV,VV,VH
HRSID [47]	SAR	-	5604	0.5m, 1 m, 3 m	HH,HV,VV,VH
NWPU-10	Optical	Google Earth	800	-	-

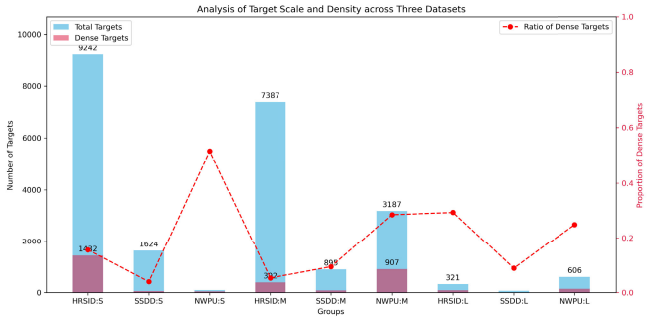


Fig. 9. Analysis chart of target scale and density in different datasets. The left and right vertical axes represent the total number of anchors and the number of dense anchors, respectively. The horizontal axis represents the scale of the targets, where S, M, and L correspond to small, medium, and large-scale types, respectively.

experiments, we investigate the prerequisites necessary for conducting this experiment on the three datasets. Specifically, we conduct statistical analyses on small, medium, and large-sized and dense targets across the three datasets. The statistical data results are illustrated in Fig. 9.

Observations reveal that, in the HRSID [47], SSDD [46], and NWPU-10 [48] datasets, the proportion of small and medium-sized targets is notably high. They account for approximately 98.1%, 97.4%, and 84.4% of the entire dataset. In addition, dense targets constitute 11.3%, 6.0%, and 28.5% of the datasets. The accuracy of remote sensing detection is often significantly influenced by medium- and small-sized targets, as well as dense small targets. The reason is that these targets tend to suffer from issues such as insufficient information extraction and ambiguous regression, which lead to network misdetection problems. Therefore, experiments conducted on the three datasets can effectively demonstrate the reliability of the experimental results and the robustness of the network.

B. Experimental Setup and Evaluation Metrics

The experiments in this study are conducted using the PyTorch deep learning framework, and both training and testing are performed in a GPU environment. For specific environment configurations, please refer to Table II. We comprehensively evaluate the performance of the remote sensing detection model using mAP, precision (P), recall (R), miss rate, and $F1$-score ($F1$) as evaluation metrics. Please refer to Table III for the meaning of each evaluation metric.

In the subsequent experiments, we set the initial learning rate of the target detection network to 0.01, momentum to 0.937, and batch size to 16. We conduct 100 iterations of training on the SSDD [46] dataset, and 300 iterations on the NWPU-10 [48] and HRSID [47] datasets. The weights obtained from training are evaluated on the test sets, and

TABLE II
EVALUATION METRICS TABLE

Evaluation index	Meaning
mAP	mean Average Precision
P	Precision (P) $= \frac{TP}{TP+FP}$. FP is False Positive.
R	Recall (R) $= \frac{TP}{TP+FN}$. FN is False Negative.
Miss Rate	$1 - \frac{TP}{TP+FN}$. TP is True Positive.
F1	$F1 = 2 \times \frac{P \times R}{P+R}$

TABLE III
EXPERIMENTAL ENVIRONMENT CONFIGURATION TABLE

Hardware and software configuration	Environment setup
Operating System	Ubuntu 20.04
CPU	Xeon(R) Silver 4210R CPU
GPU	[GeForce RTX 2080 Ti] * 4
Memory	64GB
Python version	3.8
CUDA version	11.3

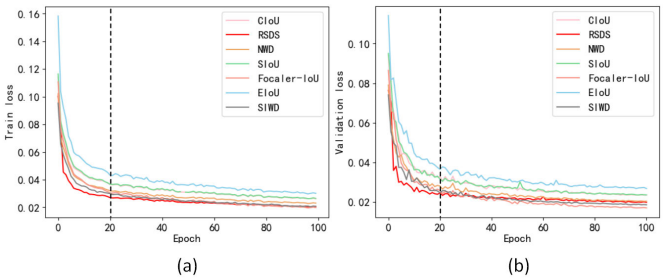


Fig. 10. Comparison of training and validation data for different regression methods on HRSID. (a) Comparison chart of training losses for different regression methods. (b) Comparison chart of validation losses for different regression methods.

the average of three trials is taken as the final result for comparative analysis of experimental outcomes.

C. Ablation Study

Training speed is compromised due to the extensive matrix operations involved in the proposed R_{NWD}Loss function, which models the similarity of distributions by Gaussian distribution modeling of 2-D detection boxes, including numerous covariance matrix calculations. We adopt a combined approach of IoU and R_{NWD} to compute regression loss for enhancing training speed. This method preserves the advantages of both loss functions while ensuring training efficiency. Through experimentation on the HRSID dataset, using the calculation formula for lbox in Algorithm 1, we determine the optimal values of γ and η, as shown in Table IV. When $\gamma = 8$ and $\eta = 2$, the model demonstrates the best mAP and $F1$ performance, while it maintains efficient time utilization during model training. Notably, adjustments can be made to the hyperparameters γ and η for different datasets. In the subsequent experiments, the baseline uses the YOLOv5 network. γ and η are set to 8 and 2, respectively, across the three remote sensing datasets.

1) Ablation Study on HRSID: We conducted experiments on the HRSID remote sensing detection dataset [47] to assess the impact of different regression loss functions on model

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TABLE IV
COMPARISON TABLE OF EVALUATION METRICS WITH VARYING η VALUES

Method	γ	η	Loss_{reg}	$\text{Loss}_{\text{inter}}$	$\text{mAP}_{0.5}$	$\text{mAP}_{0.5:0.95}$	P	R	F1
Baseline	10	0	CIoU [5]		86.72	54.41	0.8702	0.8044	0.8360
	8	2	R_{NWD}Loss	✓	92.33	64.58	0.9085	0.8739	0.8909
	7	3	R_{NWD}Loss	✓	92.26	64.32	0.9001	0.8699	0.8847
	5	5	R_{NWD}Loss	✓	91.27	64.19	0.9089	0.8524	0.8797
	3	7	R_{NWD}Loss	✓	91.40	64.35	0.9030	0.8739	0.8882
	2	8	R_{NWD}Loss	✓	92.03	63.85	0.9069	0.8734	0.8898
	0	10	R_{NWD}Loss	✓	90.11	59.47	0.9225	0.8169	0.8665

TABLE V
EXPERIMENTAL COMPARISON TABLE VALIDATING THE EFFECTIVENESS OF RSDS LOSS ON HRSID

Method	Loss_{cls}(GR)	Loss_{obj}(GR)	Loss_{reg}	$\text{Loss}_{\text{inter}}$	$\text{mAP}_{0.5}$	$\text{mAP}_{0.5:0.95}$	P	R	F1
Baseline			CIoU [5]		86.72	54.41	0.8702	0.8044	0.8360
RSDS loss	✓	✓	CIoU [5]		90.33(+3.61)	59.48(+5.07)	0.9172	0.8239	0.8681(+0.0321)
			EIoU [6]		88.87	58.42	0.8796	0.8239	0.8508
	✓	✓	EIoU [6]		89.40(+0.53)	59.05(+0.63)	0.9119	0.8120	0.8591(+0.0083)
			SIoU [7]		89.38	59.52	0.9103	0.8137	0.8593
	✓	✓	SIoU [7]		90.00(+0.62)	59.53(+0.01)	0.9167	0.8278	0.8700(+0.0107)
			Focaler-IoU [10]		89.59	59.08	0.9226	0.8105	0.8629
	✓	✓	Focaler-IoU [10]		90.57(+0.98)	60.07(+0.99)	0.9152	0.8320	0.8716(+0.0087)
			NWD [8]		90.13	59.11	0.9006	0.8316	0.8647
	✓	✓	NWD [8]		90.16(+0.03)	59.58(+0.47)	0.8972	0.8317	0.8632(-0.0015)
			SIWD [43]		90.05	59.37	0.9142	0.8235	0.8665
	✓	✓	SIWD [43]		90.32(+0.27)	59.43(+0.06)	0.9270	0.8143	0.8670(+0.0005)
			R_{NWD}Loss	✓	92.33	64.58	0.9085	0.8739	0.8909
	✓	✓	R_{NWD}Loss	✓	93.40(+1.07)	65.93(+1.35)	0.9319	0.8749	0.9025(+0.0116)

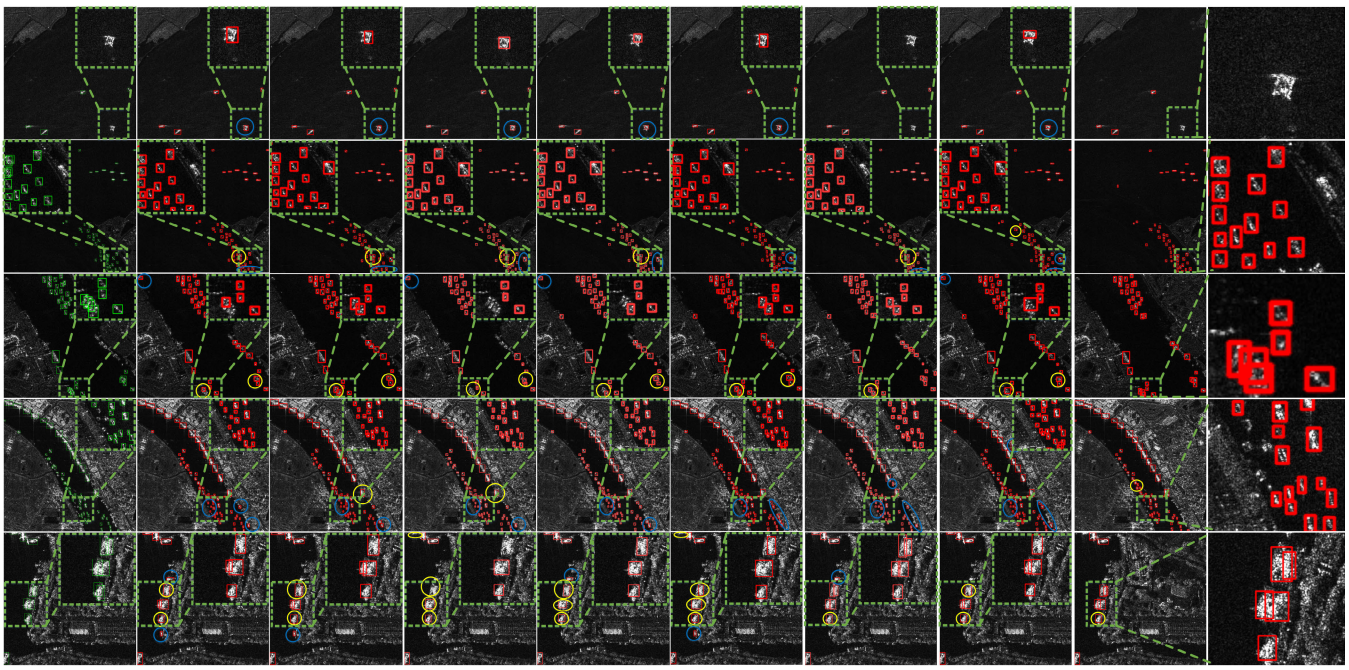


Fig. 11. Comparative illustration of detection results for DSOs across different models on the HRSID [47]. (a) GT of the images; (b) displays the detection results of the original YOLOv5 network; (c)–(h) exhibit the detection results using GRL based on different regression loss functions (CIoU [5], EIoU [6], SIoU [7], Focaler-IoU [10], NWD [8], and SIWD [43]), respectively; (i) demonstrates the detection results utilizing the RSDS loss function; and (j) showcases the region magnification effect of the images in (i). Blue circles indicate areas of incorrect detection, while yellow circles represent missed detection areas.

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TABLE VI
EXPERIMENTAL COMPARISON TABLE VALIDATING THE EFFECTIVENESS OF RSDS LOSS ON NWPU-10 DATASET

Method	Loss_{cls}(GR)	Loss_{obj}(GR)	Loss_{reg}	$\text{Loss}_{\text{inter}}$	$\text{mAP}_{0.5}$	$\text{mAP}_{0.5:0.95}$	P	R	F1
Baseline			CIoU [5]		90.05	54.79	0.9336	0.8546	0.8924
RSDS loss	✓	✓	CIoU [5]		90.12(+0.07)	54.93(+0.14)	0.9490	0.8329	0.8872(-0.0052)
			EIoU [6]		89.54	55.46	0.9179	0.8635	0.8899
	✓	✓	EIoU [6]		89.88(+0.34)	55.75(+0.29)	0.9384	0.8607	0.8979(+0.0080)
			SIoU [7]		89.21	55.14	0.9120	0.8660	0.8884
	✓	✓	SIoU [7]		89.86(+0.65)	55.98(+0.84)	0.9316	0.8541	0.8912(+0.0028)
			Focaler-IoU [10]		89.11	53.95	0.9527	0.8245	0.8840
	✓	✓	Focaler-IoU [10]		91.44(+2.33)	55.96(+2.01)	0.9195	0.8842	0.9015(+0.0175)
			NWD [8]		89.82	53.39	0.9192	0.8416	0.8787
	✓	✓	NWD [8]		90.81(+0.99)	54.33(+0.94)	0.9116	0.8690	0.8898(+0.0111)
			SIWD [43]		89.89	54.34	0.9502	0.8493	0.8969
	✓	✓	SIWD [43]		90.45(+0.56)	53.08(-1.26)	0.9537	0.8447	0.8959(-0.0010)
			R_{NWD}Loss	✓	95.14	63.05	0.9344	0.9394	0.9369
	✓	✓	R_{NWD}Loss	✓	95.31(+0.17)	63.50(+0.45)	0.9427	0.9381	0.9404(+0.0035)

performance for validating the effectiveness of the proposed R_{NWD}Loss function. In addition, we performed ablation experiments on confidence and classification losses within these models. The experimental results are presented in Table V, where values within parentheses with a "+" indicate an increase, while those with a "−" denote a decrease. In these experiments, the variables for confidence and classification losses were based on the sample Gaussian redistribution method proposed in this study. Notably, the R_{NWD} and $\text{Loss}_{\text{inter}}$ functions in this study are used in pairs, given that they both employ NWD in the loss calculation process.

By comparing the experimental data in Table V, models utilizing the Gaussian redistribution method [Loss_{cls}(GR), Loss_{obj}(GR)] are found to exhibit overall positive growth in mAP, P, R, and $F1$ values. Among the IoU-based loss functions, Focaler-IoU [10] demonstrates the best performance. After adopting the sample Gaussian redistribution method, its performance improves by 0.98% and 0.99% on $\text{mAP}_{0.5}$ and $\text{mAP}_{0.5:0.95}$, respectively. This result indicates that difficult samples receive higher weights, which allows the network to focus more on challenging instances during training. However, its P metric decreases by 0.0074, which suggests a higher proportion of misclassified positive samples. For similarity-based loss functions, the model demonstrates higher detection accuracy for small objects compared to IoU-based losses, further validating the effectiveness of similarity differences in regression loss for small object detection.

When employing the R_{NWD}Loss and $\text{Loss}_{\text{inter}}$ in our models, optimal performance is achieved, with improvements of 1.07% and 1.35% on $\text{mAP}_{0.5}$ and $\text{mAP}_{0.5:0.95}$, respectively. P, R, and $F1$ metrics each increase by 0.0234, 0.001, and 0.0116. These results indicate that the use of R_{NWD} and $\text{Loss}_{\text{inter}}$ losses not only effectively enhances the detection accuracy of the model but also improves its robustness while ensuring precision.

We conducted a performance comparison across various models, and as shown in Fig. 10, the curve of the RSDS loss drops rapidly within the first 20 epochs, indicating that the model using RSDS loss achieves the fastest convergence during this phase. Furthermore, the validation loss also shows a rapid convergence trend, with a minimal gap between the validation and training losses, suggesting that the model has achieved an optimal balance between convergence speed and stability. Subsequently, we selected images containing densely packed small objects from the test set and evaluated them using the trained models. The test results are shown in Fig. 11, while models employing GRL enhance detection performance to some extent, they still exhibit numerous instances of misdetection and false alarms, particularly in the detection of dense targets. However, with the introduction of the R_{NWD} and $\text{Loss}_{\text{inter}}$ losses, significant improvements are observed in reducing false alarms and misdetections. More importantly, targets with dense occlusion can be more effectively distinguished.

2) Ablation Study on NWPU-10: We conducted ablation experiments on the NWPU-10 dataset [48] using the same methodology, and the experimental results are presented in Table VI. In models based on IoU regression loss, the inclusion of GRL results in significant improvements in mAP and $F1$ metrics. Particularly for networks utilizing the latest Focaler-IoU [10] regression loss, $\text{mAP}_{0.5}$ and $\text{mAP}_{0.5:0.95}$ are improved by 2.33% and 2.01%, respectively, with an $F1$ value of 0.9015. This finding represents a performance increase of 0.0175 compared with networks without GRL. For networks based on WD loss, GRL effectively differentiates between sample positivity and difficulty levels, which leads to enhancements in mAP and $F1$ metrics, as detailed in the last four rows of Table VI.

However, IoU-based models still exhibit a relatively high false alarm rate for tiny objects in images, as depicted in the zoomed-in view images corresponding to the first and second rows of data in Fig. 12. In addition, the phenomenon of missed detections for dense small targets is also pronounced, as illustrated in the zoomed-in view images corresponding to the third and fourth rows of data in Fig. 10. Nevertheless, upon introducing the RSDS loss function proposed in this study, the models achieve the highest detection accuracy of 95.31% for $\text{mAP}_{0.5}$ and 63.50% for $\text{mAP}_{0.5:0.95}$, with an $F1$ score reaching its maximum value of 0.9404. This outcome results in an optimal balance between precision and recall. Moreover, models incorporating the RSDS loss function exhibit significant improvements in detecting different categories of dense

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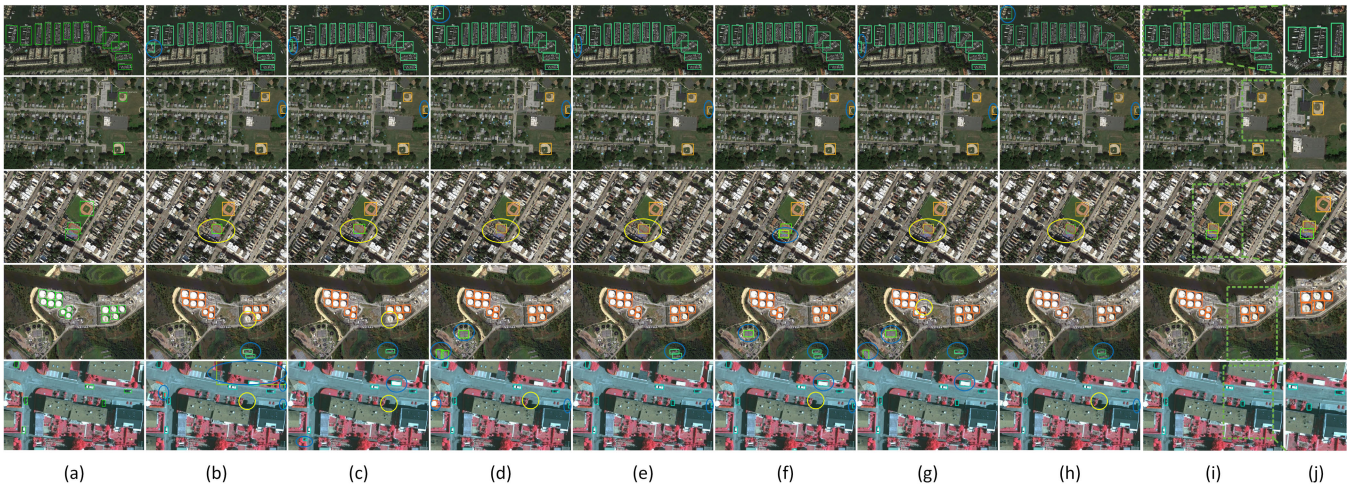


Fig. 12. Comparative illustration of detection results for DSOs across different models on the NWPU-10 dataset [48]. (a) GT of the images; (b) displays the detection results of the original YOLOv5 network; (c)–(h) exhibit the detection results using GRL based on different regression loss functions (CIoU [5], EIoU [6], SIoU [7], Focaler-IoU [10], NWD [8], and SIWD [43]), respectively; (i) demonstrates the detection results utilizing the RSDS loss function; and (j) showcases the region magnification effect of the images in (i).

TABLE VII
EXPERIMENTAL COMPARISON TABLE VALIDATING THE EFFECTIVENESS OF RSDS LOSS ON SSDD

Method	Loss_{cls}(GR)	Loss_{obj}(GR)	Loss_{reg}	$\text{Loss}_{\text{inter}}$	$\text{mAP}_{0.5}$	$\text{mAP}_{0.5:0.95}$	P	R	F1
Baseline			CIoU [5]		93.70	62.90	0.8842	0.8859	0.8850
RSDS loss	✓	✓	CIoU [5]		97.92(+4.22)	66.55(+3.65)	0.9767	0.9194	0.9472(+0.0622)
			EIoU [6]		97.57	65.87	0.9677	0.9317	0.9494
	✓	✓	EIoU [6]		97.69(+0.12)	66.50(+0.63)	0.9713	0.9304	0.9504(+0.0010)
			SIoU [7]		97.79	65.05	0.9480	0.9377	0.9428
	✓	✓	SIoU [7]		97.90(+0.11)	66.05(+1.00)	0.9864	0.9302	0.9575(+0.0147)
			Focaler-IoU [10]		97.19	65.22	0.9462	0.9377	0.9419
	✓	✓	Focaler-IoU [10]		97.66(+0.47)	65.43(+0.21)	0.9583	0.9286	0.9432(+0.0013)
			NWD [8]		97.64	65.03	0.9695	0.9304	0.9495
	✓	✓	NWD [8]		98.10(+0.46)	65.76(+0.73)	0.9748	0.9193	0.9462(-0.0033)
			SIWD [43]		98.09	70.53	0.9809	0.9432	0.9617
	✓	✓	SIWD [43]		98.36(+0.27)	70.58(+0.05)	0.9774	0.9505	0.9638(+0.0021)
			R_{NWD}Loss	✓	98.58	72.24	0.9723	0.9650	0.9686
	✓	✓	R_{NWD}Loss	✓	98.87(+0.29)	72.24(+0.00)	0.9759	0.9650	0.9704(+0.0018)

small targets, as evidenced by the detailed effects shown in the zoomed-in view images corresponding to the third row of data in Fig. 12.

3) Ablation Study on SSDD: Ablation experiments comparing models utilizing different loss functions were conducted on the SSDD dataset [46], and the results are presented in Table VII. Models incorporating GRL achieve performance improvements across metrics such as $\text{mAP}_{0.5}$, $\text{mAP}_{0.5:0.95}$, and the comprehensive evaluation metric $F1$, which combines P and R. However, for models employing NWD regression loss, although the introduction of GRL results in an improvement in mAP, the $F1$ metric decreases by 0.0033. This result is primarily attributed to the inadequate performance of NWD loss in detecting large and medium-sized objects in the original model.

Our proposed RSDS loss overcomes this issue, which enables networks incorporating this loss to excel in multiscale dense object detection. The last row of data in Table VII demonstrates that the comprehensive loss function for remote sensing DSO detection proposed in this study significantly enhances the mAP of the model while also improving its P, R, and $F1$ metrics.

The detection results of each model are illustrated in Fig. 13. For dense objects of different scales, models incorporating the RSDS loss function exhibit the best detection accuracy. Detailed effects can be observed in the zoomed-in view images in Fig. 13(j).

D. Experimental Comparison of RSDS Loss Applied to Other Object Detection Models

The experimental data above indicates that embedding the RSDS loss function proposed in this study into different detection models achieves optimal performance in DSO detection. To further validate the performance of the RSDS loss, experiments were conducted across various detection models. For anchor-based models, we replaced the regression loss function with the $\text{Loss}_{\text{inter}}$ function and GRL. For anchor-free [24], [32], [53] models such as YOLOv9 [24], while maintaining other loss functions unchanged, we modified the simple IoU calculation method to use the R_{NWD} calculation method, and simultaneously introduced GRL and $\text{Loss}_{\text{inter}}$.

We conducted experiments using state-of-the-art end-to-end detection models, and the final experimental data are

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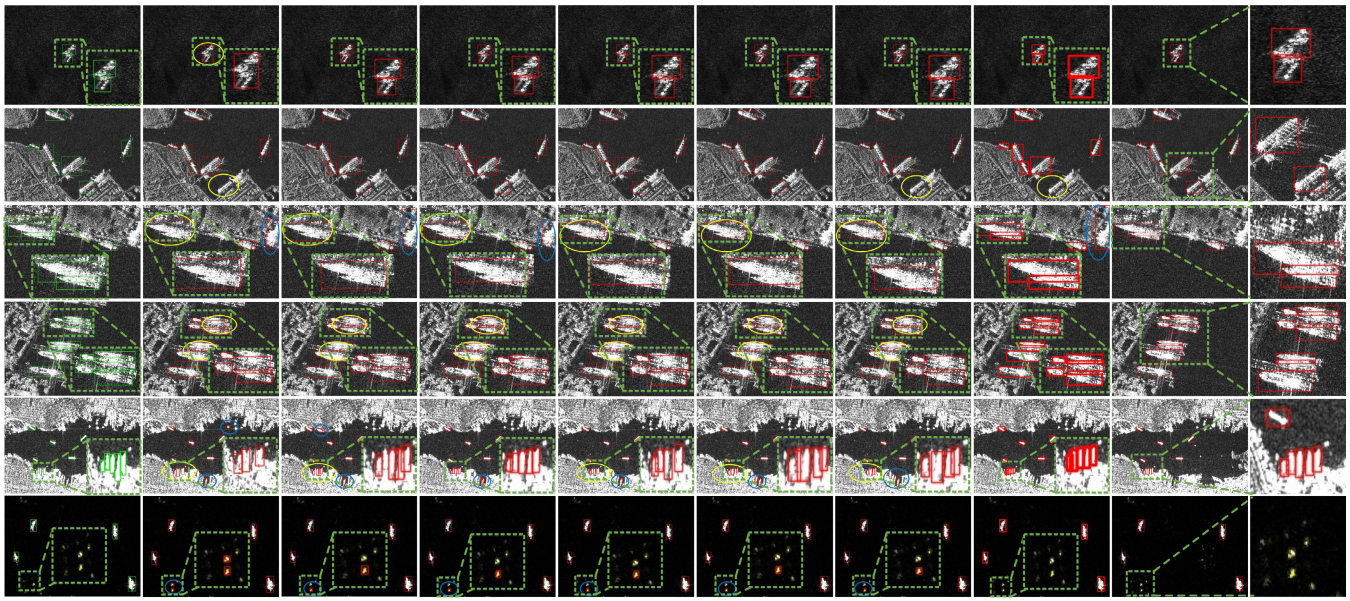


Fig. 13. Comparative illustration of detection results for DSOs across different models on the SSDD dataset [46]. (a) GT of the images; (b) displays the detection results of the original YOLOv5 network; (c)–(h) exhibit the detection results using GRL based on different regression loss functions (CIoU [5], EIoU [6], SIoU [7], Focaler-IoU [10], NWD [8], and SIWD [43]), respectively; (i) demonstrates the detection results utilizing the RSDS loss function; and (j) showcases the region magnification effect of the images in (i).

TABLE VIII
EXPERIMENTAL COMPARISON TABLE VALIDATING THE EFFECTIVENESS OF RSDS LOSS ON HRSID

Method	Backbone	$mAP_{0.5}$	$mAP_{0.5:0.95}$	P	R	Miss Rate	F1
YOLOv5	CSPDarknet53 [15]	86.72	54.41	0.8702	0.8044	0.1956	0.8360
YOLOv5*	CSPDarknet53 [15]	93.40(+6.70)	65.93(+11.49)	0.9319	0.8749	0.1251	0.9025(+0.0665)
YOLOv7 [11]	RepVGG [11]	87.44	54.23	0.8777	0.7829	0.2171	0.8276
YOLOv7*	RepVGG [11]	88.60(+1.16)	59.35(+5.12)	0.8871	0.8200	0.1800	0.8522(+0.0246)
SWIN-trans [15], [49]	Swin-transformerv2 [49]	89.70	59.47	0.9075	0.8204	0.1796	0.8618
SWIN-trans*	Swin-transformerv2 [49]	92.94(+3.24)	64.91(+5.44)	0.9272	0.8646	0.1354	0.8948(+0.0330)
PP-PicoDet [50]	ESNet [50]	86.75	56.34	0.9090	0.7678	0.2322	0.8325
PP-PicoDet*	ESNet [50]	87.88(+1.13)	56.44(+0.10)	0.8987	0.7890	0.2110	0.8403(+0.0078)
CS^nNet [51]	Hornet [52]	88.42	57.71	0.8990	0.8067	0.1933	0.8504
CS^nNet*	Hornet [52]	90.58(+2.16)	59.87(+2.16)	0.9201	0.8158	0.1842	0.8648(+0.0144)
YOLOv9 [24]	GELAN [24]	89.63	62.72	0.9125	0.7998	0.2002	0.8524
YOLOv9*	GELAN [24]	91.42(+1.79)	65.10(+2.38)	0.9126	0.8314	0.1686	0.8701(+0.0177)

presented in Table VIII. The results indicate that YOLOv5* exhibits the best performance across all evaluation metrics, with performance improvements of 6.70% and 11.49% in $mAP_{0.5}$ and $mAP_{0.5:0.95}$, respectively, and an increase of 0.0612 in $F1$ score. We also explored a method called Swin-trans [15], [49], which employs Swin-transformerv2 [49] as the backbone network for feature extraction while retaining the neck and head parts of YOLOv5. Although Swin-trans shows improvements in all evaluation metrics after using the RSDS loss, its performance does not reach the level of the YOLOv5* model. Notably, YOLOv9* achieves a significant performance improvement of 0.83% in $mAP_{0.5:0.95}$, which surpasses the optimal result. Therefore, this method is suitable for subsequent DSO detection tasks.

V. CONCLUSION AND FUTURE WORK

This study proposes a novel RSDS loss function to address the issues of occlusion and missed detections commonly encountered with DSOs in RSI. The loss function consists of GRL, R_{NWD}Loss, and $\mathrm{Loss}_{\mathrm{inter}}$. GRL utilizes adaptive weighting factors to dynamically redistribute sample weights, thus enhancing detection performance. The R_{NWD}Loss models the qualified 2-D detection boxes as Gaussian distributions and measures the similarity between the target boxes using Wasserstein distance (WD). This approach addresses the issue of zero loss between nonintersecting targets, a limitation that traditional IoU and IoG metrics cannot handle. Finally, $\mathrm{Loss}_{\mathrm{inter}}$ considers attraction between dense targets and repulsion from surrounding objects, which enables candidate boxes to more accurately match their GT boxes in complex environments. Experimental results demonstrate that the RSDS loss significantly reduces the probability of missed detections for remote sensing small objects and performs outstandingly on the HRSID, NWPU-10, and SSDD datasets. The optimal detection models, after adopting the RSDS loss, achieve improvements of 6.68% and 11.52%, 5.26% and 8.71%, and 5.17% and

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9.34% in $mAP_{0.5}$ and $mAP_{0.5:0.95}$ compared with baselines. In addition, the RSDS loss can be embedded into anchor-free and anchor-based models to enhance model performance.

In future work, we will integrate our RSDS loss into the task of remote sensing rotation object tracking to accurately capture dense small vessel targets in scenarios such as ports and traffic hubs. By analyzing their historical information and predicting their trajectories in occluded videos, we aim to further contribute to the perception tasks of intelligent vessels.

APPENDIX

A. Proof From (8) to (9)

Given

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|^2 + \mathrm{Tr}\left(\Sigma_1 + \Sigma_2 - 2\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right).$$

Since

$$\mathrm{Tr}\left(\Sigma_1 + \Sigma_2 - 2\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right) = \mathrm{Tr}(\Sigma_1) + \mathrm{Tr}(\Sigma_2) - 2\mathrm{Tr}\left(\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right).$$

And considering that Σ_1 and Σ_2 are both symmetric matrices

$$\mathrm{Tr}\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right) = \mathrm{Tr}(\Sigma_1\Sigma_2).$$

Therefore,

$$\mathrm{Tr}(\Sigma_1) + \mathrm{Tr}(\Sigma_2) - 2\mathrm{Tr}\left(\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right) = \mathrm{Tr}(\Sigma_1) + \mathrm{Tr}(\Sigma_2) - 2\mathrm{Tr}(\Sigma_1\Sigma_2)^{1/2}.$$

Which can be written as

$$= \mathrm{Tr}\left(\left(\Sigma_1^{1/2} - \Sigma_2^{1/2}\right)^2\right) = \|\Sigma_1^{1/2} - \Sigma_2^{1/2}\|^2_{\text{Frobenius}}.$$

Finally, we have

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|^2 + \|\Sigma_1^{1/2} - \Sigma_2^{1/2}\|^2_{\text{Frobenius}}.$$

B. Proof of $W_2^2(\mathcal{N}_a, \mathcal{N}_b)$

The squared ED between vectors can be calculated by finding the square of the difference between the vectors

$$W_2^2(\mathcal{N}_a, \mathcal{N}_b) = \|V_a - V_b\|_2^2.$$

Expanding the Euclidean norm

$$= (V_a - V_b)^T(V_a - V_b).$$

Breaking down the vectors into their components

$$= (C_{xa} - C_{xb})^2 + (C_{ya} - C_{yb})^2 + \left(\frac{W_a}{2} - \frac{W_b}{2}\right)^2 + \left(\frac{H_a}{2} - \frac{H_b}{2}\right)^2.$$

Here, C_{xa} and C_{xb} are the x-coordinates, C_{ya} and C_{yb} are the y-coordinates, W_a and W_b are the widths, and H_a and H_b are the heights of the vectors V_a and V_b, respectively.

C. Simplified Algorithm for R_{NWD}Loss

Algorithm 2 Sample R_{NWD}Loss

Require: box1, box2, $\text{reg}_{\text{weight}} = 0.01$
Ensure: R_{NWD}Loss

box2 = box2^T
$b1_{\text{cx}}, b1_{\text{cy}}, b1_{\text{w}}, b1_{\text{h}}$ = box1[0], box1[1], box1[2], box1[3]
$b2_{\text{cx}}, b2_{\text{cy}}, b2_{\text{w}}, b2_{\text{h}}$ = box2[0], box2[1], box2[2], box2[3]
$\text{valid}_{\text{indices}} = \text{anchor}_{\text{area}} < \Theta$
if torch.any($\text{valid}_{\text{indices}}$) **then**
  Compute squared Euclidean distance between centers: $p1 = (b1_{\text{cx}} - b2_{\text{cx}})^2 + (b1_{\text{cy}} - b2_{\text{cy}})^2$
  Compute squared Euclidean distance between width and height: $p2 = \left(\frac{b1_{\text{w}} - b2_{\text{w}}}{2}\right)^2 + \left(\frac{b1_{\text{h}} - b2_{\text{h}}}{2}\right)^2$
  Compute regularization term: $\text{reg}_{\text{term}} = \text{reg}_{\text{weight}} \times (b1_{\text{cx}}^2 + b1_{\text{cy}}^2 + b1_{\text{w}}^2 + b1_{\text{h}}^2)$
  Compute loss with regularization: $R_{\text{NWD}} = p1 + p2 + \text{reg}_{\text{term}}$
else
  return IoU
end if
Update lbox:
lbox$+= \gamma \times (1.0 - \text{IoU.mean}()) + \eta \times (1.0 - R_{\text{NWD}}.\text{mean}())$

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Chengcheng Chen (Graduate Student Member, IEEE) received the bachelor's degree in computer science and technology from Jining University, Qufu, Shandong, China, in 2020. He is currently pursuing the Ph.D. degree with the Merchant Marine College, Shanghai Maritime University, Shanghai, China.

His research interests mainly include small object detection, semantic segmentation, and object tracking.



Weiming Zeng (Senior Member, IEEE) received the Ph.D. degree in biomedical engineering from Southeast University, Nanjing, China, in 2005.

He is currently a Full Professor with the College of Information Engineering, Shanghai Maritime University, Shanghai, China. His research interests include image processing, analyzing functional connectivity, and neuroscience computation.

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Xiliang Zhang (Graduate Student Member, IEEE) received the B.S. degree in Internet of Things engineering from Qingdao University of Science and Technology, Qingdao, China, in 2020. He is currently pursuing the Ph.D. degree with the College of Information Engineering, Shanghai Maritime University, Shanghai, China.

His research interests mainly include deep learning, situational awareness of autonomous ships, and intelligent transportation systems.



Yuhao Zhou is currently pursuing the Ph.D. degree with the School of Software Engineering, Tongji University, Shanghai, China.

He has authored some academic articles which have been published in refereed international journals, such as IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS, *ACM Transactions on Design Automation of Electronic Systems*, and *Applied Soft Computing*. His research interests include Reed–Muller circuits area optimization based on XNOR/OR, power optimization, low-power integrated circuits design, computer-aided design, and intelligent optimization algorithm.

Mr. Zhou is a member of ACM.



Juan Yu received the master's degree in vehicle application engineering from Shanghai Maritime University, Shanghai, China, in 2015, where she is currently pursuing the Ph.D. degree in traffic information engineering and control.

Her research directions mainly include the application of AI visual intelligent perception analysis technology and big data analysis technology in the fields of intelligent transportation and intelligent shipping.



Yugang Chang received the bachelor's degree in naval architecture and marine engineering from Jimei University, Xiamen, Fujian, China, in 2016. He is currently pursuing the Ph.D. degree with the Merchant Marine College, Shanghai Maritime University, Shanghai, China.

His research interests mainly include object tracking, small object detection, and intelligent transportation systems.



Fei Wang received the bachelor's degree in computer science and technology from Shanghai Polytechnic University, Shanghai, China, in 2023. He is currently pursuing the master's degree with the College of Information Engineering, Shanghai Maritime University, Shanghai.

His research interests mainly include deep learning and object detection.

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